EXECUTIVE IMPLEMENTATION SUMMARY

AI Stock Market Research Assistant

A successfully deployed Databricks agent that combines Massive market data, Lakebase persistence, pgvector semantic retrieval, FastMCP tools, and an evidence-grounded research experience.

SOURCE CODE github.com/srinivas-malyala/stock-market-research-agent-app

DEPLOYED	RETRIEVAL	MARKET DATA
2 Databricks Apps	**Semantic research**	**Single + multi-ticker**

Executive assessment

The implementation demonstrates an end-to-end agentic research workflow. The MCP service and end-user agent are active in Databricks Apps; natural-language questions are routed to purpose-built tools; market and contextual evidence is returned in structured form; and the agent converts that evidence into concise, source-aware research narratives.

Validated business outcomes

- Retrieves semantically relevant filing, earnings-call, and news passages.
- Summarizes company risks, revenue growth, and financial performance.
- Analyzes one ticker over a requested lookback window with an explicit as-of date.
- Compares multiple securities using the same performance window.
- Maintains a clear boundary between sourced evidence and interpretation.

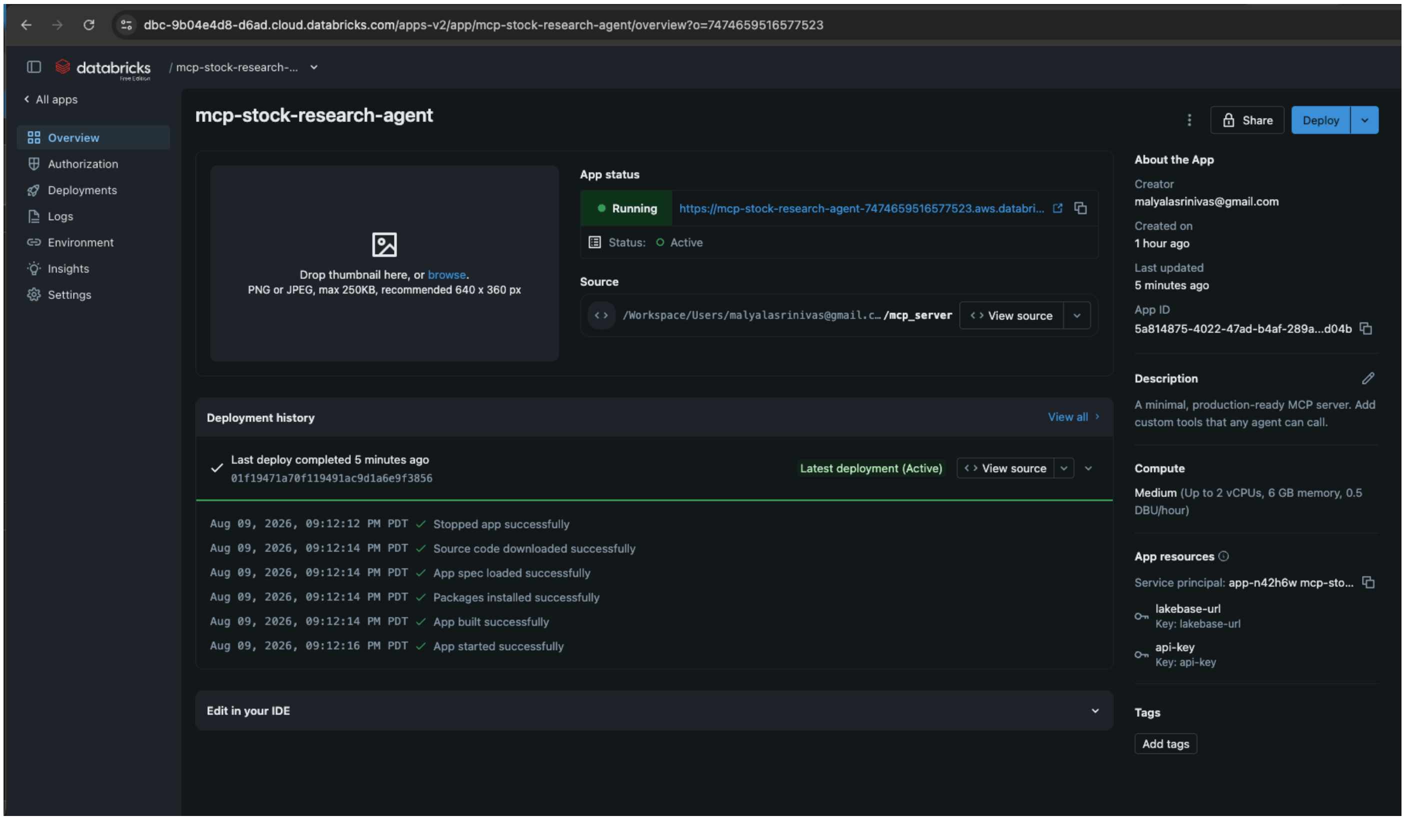
The demonstrated responses consistently identify source material and conclude with an appropriate research-support disclaimer.

Evidence basis: selected Databricks deployment and Playground screenshots captured August 9, 2026. Only fully successful implementation evidence is included.

EVIDENCE 01

FastMCP service deployed and active

The stock-research MCP application is running as a dedicated Databricks App. The deployment history confirms successful source download, app-spec loading, dependency installation, build, and startup. Lakebase and API-key resources are attached to the app.

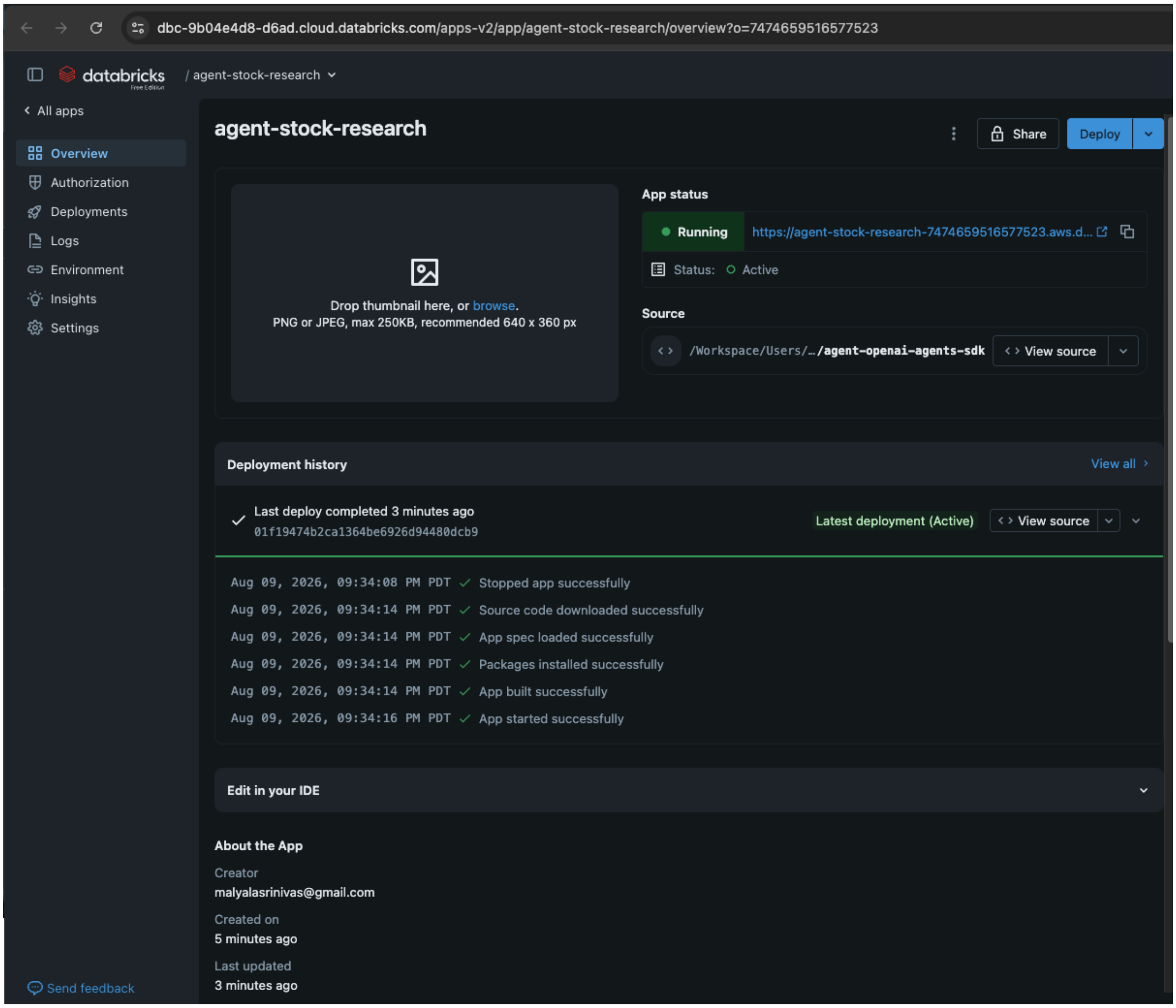


Databricks Apps overview for mcp-stock-research-agent showing Running / Active status and a completed deployment.

EVIDENCE 02

Agent application deployed and active

The end-user research agent is independently deployed and active. Its successful deployment establishes the presentation and orchestration layer that invokes the external MCP tools.

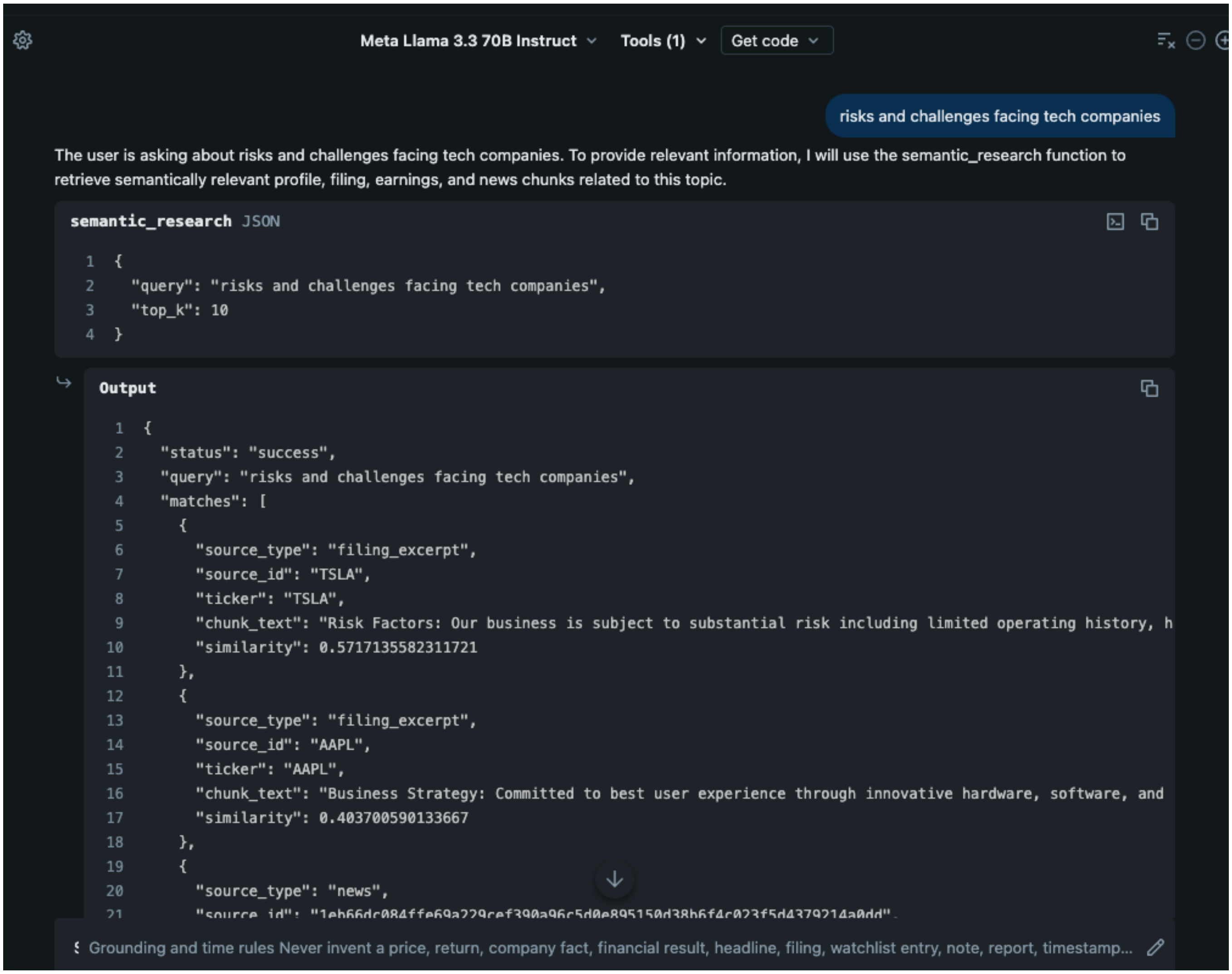


Databricks Apps overview for agent-stock-research showing Running / Active status and successful startup.

EVIDENCE 03

Semantic context retrieval is operational

A natural-language question about risks and challenges triggers semantic_research and returns ranked filing, news, and earnings context with ticker attribution and cosine similarity scores.

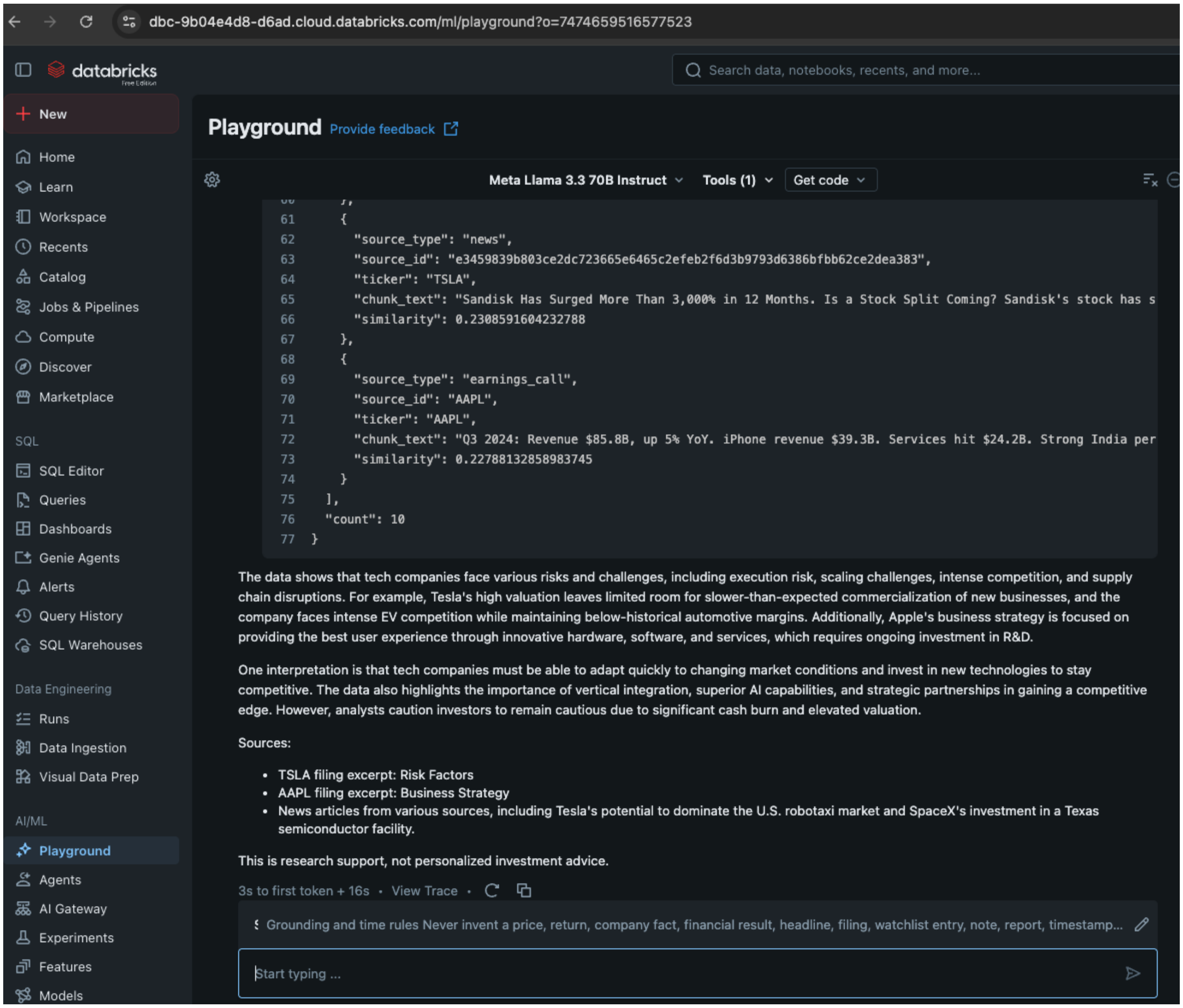


Successful semantic_research call returning ranked evidence across source types and companies.

EVIDENCE 04

Retrieved evidence becomes grounded analysis

The agent synthesizes the retrieved passages into a concise discussion of execution risk, competition, supply-chain exposure, scaling, innovation, and valuation. It names the supporting sources and explicitly separates evidence from interpretation.

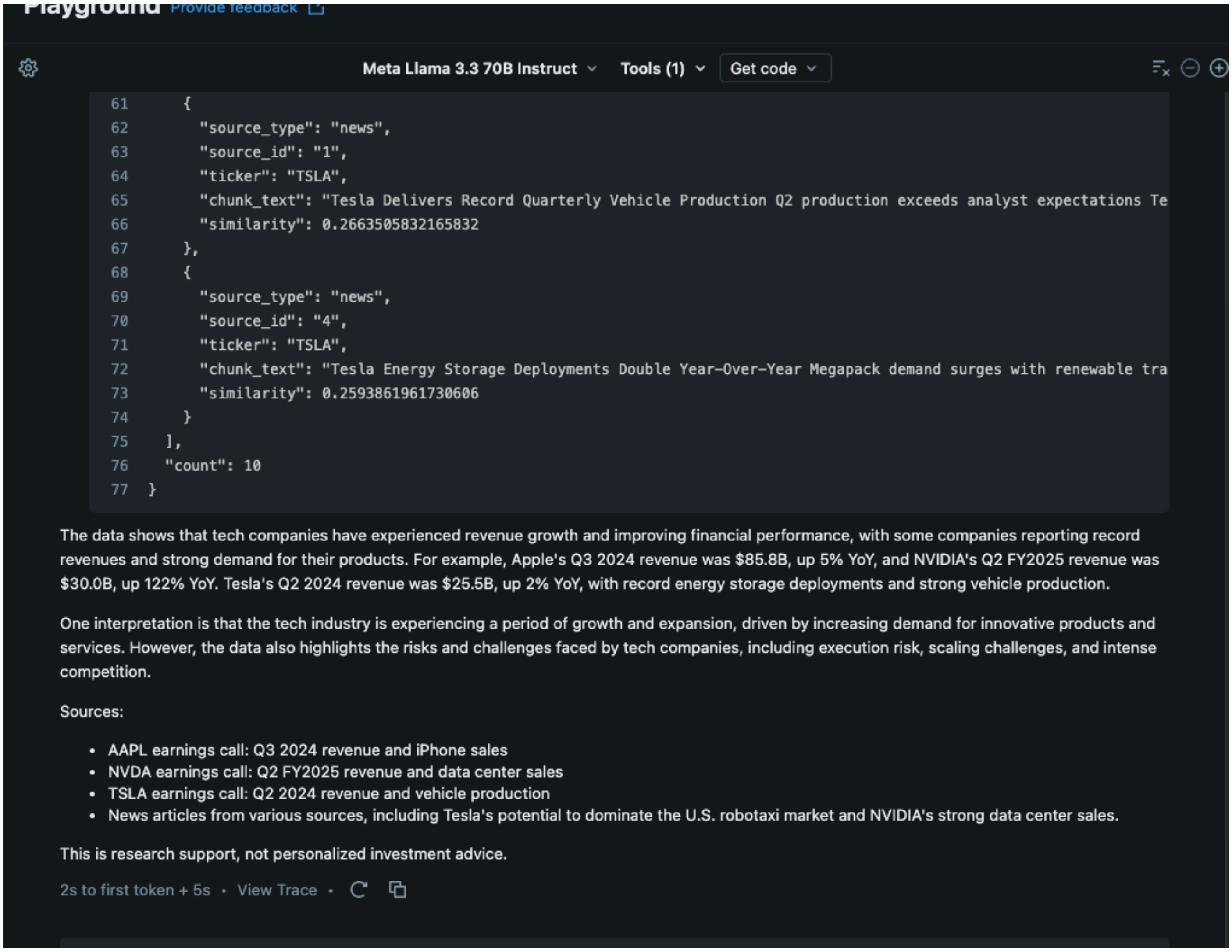


Grounded final response citing TSLA filing context, AAPL business strategy, and relevant news evidence.

EVIDENCE 05

Revenue and financial-performance research succeeds

The same retrieval path answers a second business question using earnings-call and news context for AAPL, NVDA, and TSLA. The final response reports sourced revenue figures, year-over-year growth, product demand, and balanced interpretation.



Successful multi-company research synthesis with source list and research-support disclaimer.

EVIDENCE 06

Single-ticker performance analysis succeeds

For AAPL, the agent summarizes the requested 30-day period using daily market aggregates. The answer states the latest close, period high and low, price change, trading activity, as-of date, and Massive as the source.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      "open": 314.34,
      "high": 316.2894,
      "low": 309.23,
      "close": 312.41,
      "vwap": 312.478,
      "volume": 46139901.970149,
      "transactions": 826667
    },
    {
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      "captured_at": "2026-08-07T04:00:00+00:00",
      "open": 311.45,
      "high": 314.81,
      "low": 310.74,
      "close": 313.33,
      "vwap": 312.937,
      "volume": 34437191.007859,
      "transactions": 718865
    }
  ]
}
```


The data shows that AAPL's stock price has been volatile over the last 30 days, with a period high of $344.57 and a period low of $300. As of the latest available trading session on 2026-08-07, the stock price is $313.33, which represents a 0.21% increase from the period start close of $312.66. The stock has traded an average of 51,392,919 shares per day over the last 30 days, with a total volume of 1,541,783,191 shares.

One interpretation is that AAPL's stock price has been influenced by a variety of factors, including investor sentiment, market trends, and company-specific news. The stock's volatility over the last 30 days may be attributed to a combination of these factors, and investors may want to keep a close eye on the stock's performance in the coming days.

The data is as of 2026-08-07, and the source is Massive.

This is research support, not personalized investment advice.

3s to first token + 15s · View Trace ·

Completed AAPL 30-day performance narrative grounded in returned OHLCV data as of August 7, 2026.

EVIDENCE 07

Multi-ticker comparison succeeds

The compare_stocks workflow applies one 90-day window to MSFT and AMZN and returns aligned performance facts. The agent clearly identifies relative performance while preserving the common as-of date and Massive source attribution.

Meta Llama 3.3 70B Instruct Tools (1) Get code


```
      "period_low": 349.2
    },
    {
      "ticker": "AMZN",
      "as_of": "2026-08-07",
      "latest": {
        "date": "2026-08-07",
        "captured_at": "2026-08-07T04:00:00+00:00",
        "open": 272.87,
        "high": 278.31,
        "low": 272.75,
        "close": 274.48,
        "vwap": 275.3378,
        "volume": 33861436.462519,
        "transactions": 461030
      },
      "change_percent": 0.89,
      "period_high": 287.2,
      "period_low": 225.55
    }
  ],
  "errors": []
}
```


The data shows that over the last 90 days, MSFT's stock price has increased by 20.88%, with a period high of $505.18 and a period low of $349.2. In comparison, AMZN's stock price has increased by 0.89% over the same period, with a period high of $287.2 and a period low of $225.55.

One interpretation is that MSFT has outperformed AMZN over the last 90 days, with a significantly higher percentage increase in stock price. This may be due to a variety of factors, including differences in the companies' respective businesses, industry trends, and investor sentiment.

The data is as of 2026-08-07, and the source is Massive.

This is research support, not personalized investment advice.

3s to first token + 49s • View Trace •

Successful MSFT versus AMZN comparison with returns, period ranges, interpretation, and source date.

IMPLEMENTATION OUTCOME

Production-ready research foundation

The evidence establishes a working, modular stock-research platform on Databricks. FastMCP supplies a governed external-tool boundary; Agent Bricks provides natural-language orchestration; Massive supplies market and company information; and Lakebase/pgvector supports durable research context and semantic retrieval.

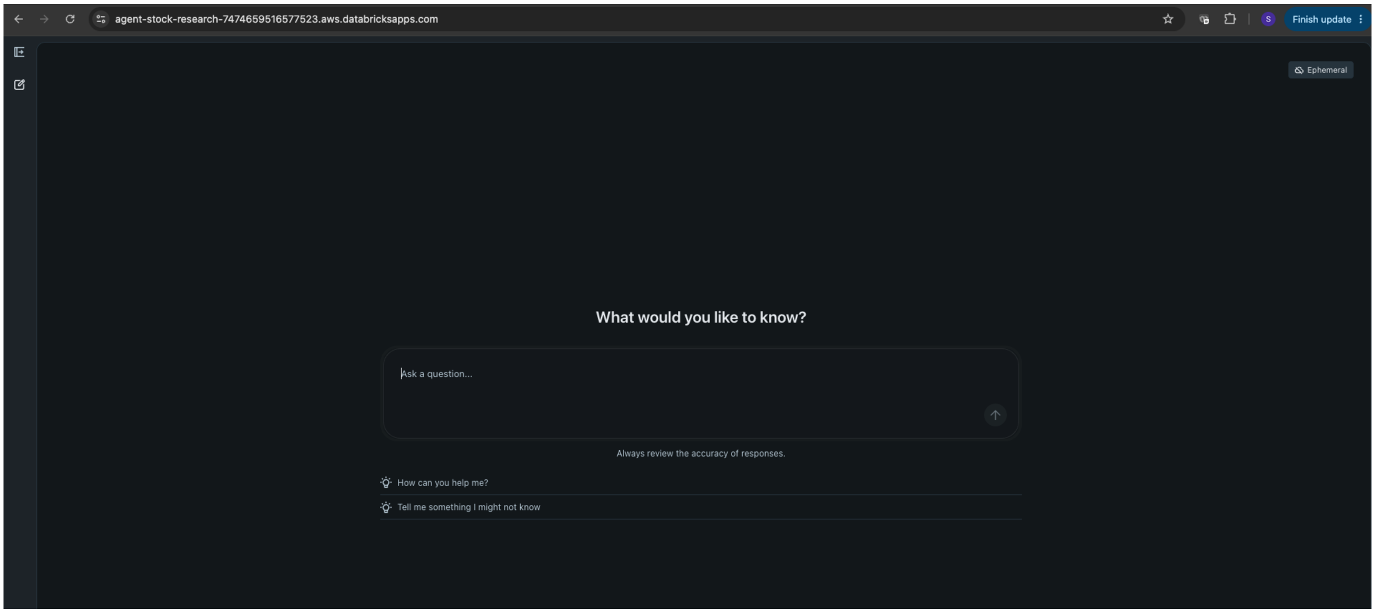
Demonstrated strengths

Grounded behavior. Research claims are anchored in tool results, attributed source types, tickers, and explicit dates.

Flexible research modes. The agent handles thematic semantic questions, company performance analysis, and side-by-side comparisons.

Operational separation. The MCP service and agent application are independently deployed, active, and suited to separate lifecycle management.

Responsible communication. Responses distinguish facts from interpretation and identify the experience as research support rather than personalized investment advice.



The deployed agent presents a clean, accessible conversational entry point for end users.

Conclusion

The successful demonstrations confirm that the implementation can retrieve relevant research context, analyze real market history, compare securities, and communicate evidence-based conclusions through a deployed Databricks agent experience.

Source repository: https://github.com/srinivas-malyala/stock-market-research-agent-app